WK 04: From Your Computer to Cloud Computing

A Fractal View of Data Infrastructure

PPOL 5206 | Massive Data Fundamentals | Spring 2026

Today's Journey

Part 1-2

The Problem & Inside the Machine

Part 3-4

Enabling Abstractions & What Cloud Is

Part 5-6

Service Models & Data Centers

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Regions & CERN Case Study

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AWS Services & Synthesis

The Data Problem

CERN's Impossible Problem

The Large Hadron Collider produces:

40M
events/second

~50 TB/s
raw data rate

~10 GB/s
storage capacity

The Question:
How do they decide what to keep? This is the fundamental problem of data infrastructure at every scale.

The Physics Constraint

Speed of Light

~1 foot

per nanosecond

NYC → London

~28ms

minimum latency

This is not a technology problem to solve, it is a constraint to manage.

The Economics Constraint

Faster = More Expensive

Faster storage technologies cost more per byte

Moving Data Costs

Time, energy, and money to transfer

Proximity Premium

Closer to compute = faster but pricier

Embedded Decisions

Every architecture reflects trade-offs

These constraints create the same patterns at every scale.

The Fractal Premise

Once you understand why your laptop's L1 cache exists, you understand why CDN edge locations exist.

Same Problem → Same Solution → Different Magnitude

CHIP		DATA CENTER		GLOBAL
L1 Cache	→	**Local NVMe**	→	**CDN Edge**
~1 ns		~100 µs		~10 ms

> "Is the cloud just someone else's computer?"
> Yes, but, the interesting part is *why* it's organized the way it is.

Judgment Question

DISCUSSION

A colleague says **"storage is basically free now, so we should just keep everything."**

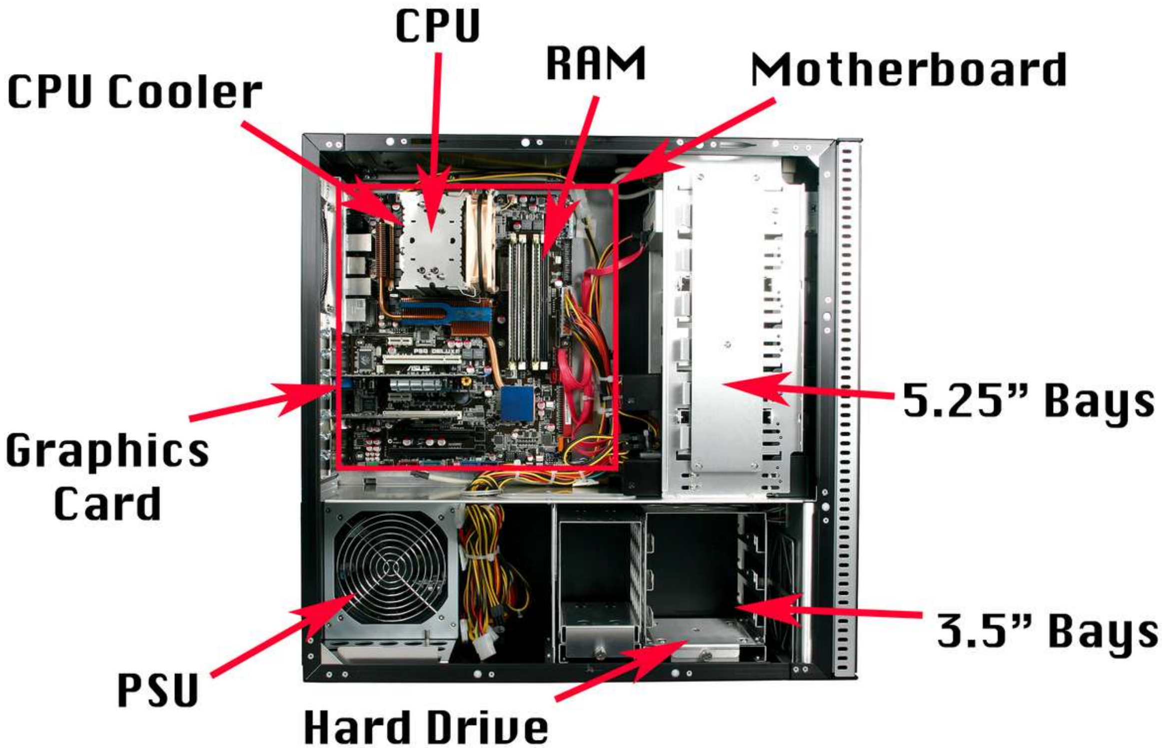
What's wrong with this reasoning?

Under what circumstances might it be approximately correct?

Inside the Machine

The “Nano” Scale

A Computer’s Parts



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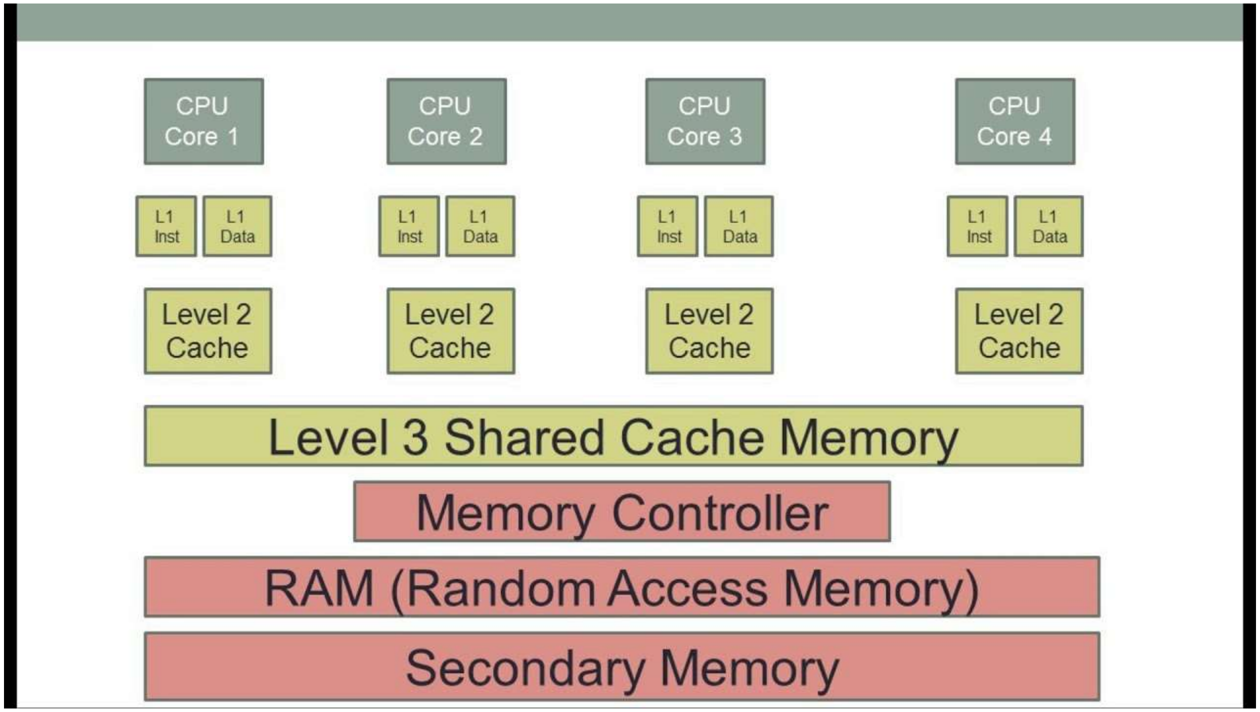
[How Does CPU Cache Work and What Are L1, L2, and L3 Cache? - The Tech Edvocate]

Memory Structrure



Computer Component Explanation

What each component does

Important for understanding the various roles in a cloud system

- Compute
- Storage
- - Etc.

Source: Citation goes here

The Memory Hierarchy

The organizing principle of computer architecture:

Level	Size	Latency	Cost/GB
CPU Registers	~1 KB	< 1 ns	N/A
L1 Cache	32-64 KB	~1 ns	Very high
L2/L3 Cache	256 KB - 64 MB	4-20 ns	High
RAM (DRAM)	16-512 GB	~100 ns	~$3-5
SSD	256 GB - 8 TB	~100 μs	~$0.10
HDD	1-20 TB	~10 ms	~$0.02

Key insight: Each level is ~10-100x slower but ~10-100x larger and cheaper.

Why Multiple Cache Levels?

Trade-off Points

Each level balances speed, size, and cost differently

Cache Misses Are Expensive

CPU must wait for slower memory

This hierarchy exists because of **physics**:

Faster memory must be physically closer to the CPU.

Storage & Specialized Compute

Storage Types

RAM (Volatile)
Fast, expensive, loses data when power lost

SSD (Non-volatile)
No moving parts, relatively fast

HDD (Non-volatile)
Spinning platters, cheap for bulk

GPUs

Thousands of simple cores optimized for parallel operations

Originally graphics; now essential for **ML, simulation, video**

Trade-off: Sacrifice single-thread performance for massive parallelism

The Pattern to Notice

Faster = Smaller + Expensive

Data Locality Matters

Specialization ↔ Flexibility

Physics Governs All

This exact pattern will reappear at data center and global scales.

Judgment Question

- Explain AWS's various CPU options and GPU Options

DISCUSSION

You're advising an agency purchasing new servers. They ask whether to prioritize **more RAM** or **larger SSDs**.

What questions would you ask about their workload before answering?

How the Cloud Became Possible

The Enabling Abstractions

The Historical Arc

Mainframes (1960s-80s)

Time-sharing: multiple users, one expensive machine. Centralized compute.

Sound familiar?

PCs (1980s-2000s)

Distributed compute to individuals. Each machine independent.

Problem: mostly idle

Internet Era

Connected machines. Always-on services demanded.

Scale problem emerged

Virtualization: The Enabling Abstraction

The Problem

Physical servers are expensive
Typical utilization: only 10-15%
Can't easily move workloads

The Solution

Hypervisor: one physical → many logical
VMs think they have dedicated hardware
Workloads become portable

Without virtualization, no multi-tenancy.
Without multi-tenancy, no cloud economics.

Types of Hypervisors

Feature	Type 1 (Bare Metal)	Type 2 (Hosted)
Installation	Directly on hardware	On top of existing OS
Performance	High, low latency	Lower due to OS overhead
Security	More secure	Increased attack surface
Use Case	Data centers, cloud	Development, testing
Examples	VMware ESXi, Hyper-V, Xen	VirtualBox, VMware Workstation

Cloud providers use Type 1 for performance and security.

VMs - Containers - Serverless

Virtual Machines

Each runs full OS
Strong isolation
Slower startup
More overhead

→

Containers

Share OS kernel
Lighter weight
Faster startup
Weaker isolation

→

Serverless

Just provide code
Infrastructure abstracted
Pay per execution
Cold start latency

The Cloud Computing Mindset

What changed wasn't just technology, it was thinking:

Specialization

Let someone else manage infrastructure

Elasticity

Scale up for demand, scale down after

Commodity Machines

Servers are disposable, replaceable

Design for Failure

Assume components will fail; architect around it

Definition of Cloud Computing

The NIST Definition

> "Cloud computing is a model for enabling ubiquitous, convenient, *on-demand network access* to a shared pool of configurable computing resources that can be *rapidly provisioned and released* with minimal management effort."
>
> — NIST Special Publication 800-145, [The NIST Definition of Cloud Computing | NIST]

Five Essential Characteristics:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

NIST: Essential Characteristics

On-demand self-service: Users can independently provision resources like server time and storage as needed, without requiring provider interaction.

Broad network access: Resources are accessible over the network via standard methods, allowing use by various devices (e.g., mobile phones, tablets, laptops).

Resource pooling: Providers use a multi-tenant model to allocate resources dynamically, offering flexibility in resource location (e.g., datacenter, state).

Rapid elasticity: Resources can be quickly scaled up or down as demand changes, appearing unlimited to the consumer.

Measured service: Usage is metered and reported, enabling optimized resource control and transparency for both provider and consumer.

Deployment Models

Model	Description	Use Case
Private	Exclusive use by single organization	Regulated industries, sensitive data
Community	Shared by organizations with common concerns	Government, research consortia
Public	Open for general public use	Most commercial applications
Hybrid	Combination of two or more types	Burst capacity, data sovereignty

CapEx to OpEx

Traditional IT (CapEx)

Buy servers, depreciate 3-5 years

Guess future capacity needs

Pay whether you use it or not

Cloud (OpEx)

Pay by hour or second

Scale up and down as needed

Variable costs track with business

Judgment Question

DISCUSSION

An agency CIO says **"we're moving to cloud to reduce costs."**

What follow-up questions would reveal whether this is likely to be true for their specific situation?

Cloud Service Models

Shared Responsibility

The Service Model Spectrum

Model	What It Provides	You Manage	Provider Manages
IaaS	VMs, storage, network	OS, runtime, app, data	Hardware, virtualization
PaaS	Dev/deployment platform	Application, data	OS, runtime, scaling
SaaS	Complete application	Data (mostly)	Everything else
Serverless	Function execution	Code, data	All infrastructure

More abstraction = less control but less operational burden

Shared Responsibility Model

Layer	On-Prem	IaaS	PaaS	SaaS
Applications	**You**	**You**	**You**	Provider
Data	**You**	**You**	**You**	Provider*
Runtime	**You**	**You**	Provider	Provider
Operating System	**You**	**You**	Provider	Provider
Virtualization	**You**	Provider	Provider	Provider
Servers/Storage/Network	**You**	Provider	Provider	Provider

Critical: "Moving to cloud" doesn't eliminate security responsibilities, it changes them. Misconfigured S3 buckets and IAM are leading breach causes.

How to Choose: Decision Framework

Consider...	IaaS	PaaS	Serverless
Control needed	High	Medium	Low
Ops expertise available	Yes	Some	Minimal
Workload pattern	Steady	Variable	Event-driven
Migration complexity	Lowest	Medium	Refactoring
Cost predictability	Higher	Medium	Variable

Judgment Question

DISCUSSION

Your agency needs to run a **Python script that processes uploaded documents**. It runs ~30 seconds per document. You expect 100-500 documents/day with unpredictable timing.

Which service model would you recommend?

What would change your answer?

Inside the Data Center

Computing at the Meso Scale

The Memory Hierarchy Reappears

Just as computers have cache → RAM → SSD → HDD, data centers have storage tiers:

Tier	Technology	Use Case	Cost
Hot	NVMe SSD, local to compute	Active queries, real-time	$$$$
Warm	SSD storage arrays	Recent data, frequent access	$$$
Cold	HDD arrays, object storage	Historical, infrequent	$$
Archive	Tape, deep archive	Compliance, rarely accessed	$

Cloud pricing makes this explicit: S3 Standard → Intelligent-Tiering → Glacier → Deep Archive

PHYSICAL SECURITY

- Separate Buildings
- Restricted Access
- Controlled environment
  - Fire suppression
  - Generators
  - Cooling
- Disaster Resilience
  - Separate regions
- High speed connections
  - Intra
  - Inter
- Disposal and Decommisioning

Networking Infrastructure

- Physical networks
  - High-speed cabling
  - Routers and switches
- Software Defined Networks
  - Network resources are virtualized
  - Made available programmatically through APIs
  - Virtual Private Clouds (VPC)
- Users must set up *PUBLIC* and *PRIVATE* interfaces
  - IP addresses
  - Subnets
  - Security Groups
  - Internal IDs: (Amazon Resource Names ARNs)

Physical Infrastructure

Power

10-100+ megawatts per facility
Redundant feeds, UPS, diesel generators
Often largest operating cost

Cooling

Every watt → heat to remove
CRAC, liquid, immersion cooling
Cooler climates = lower costs

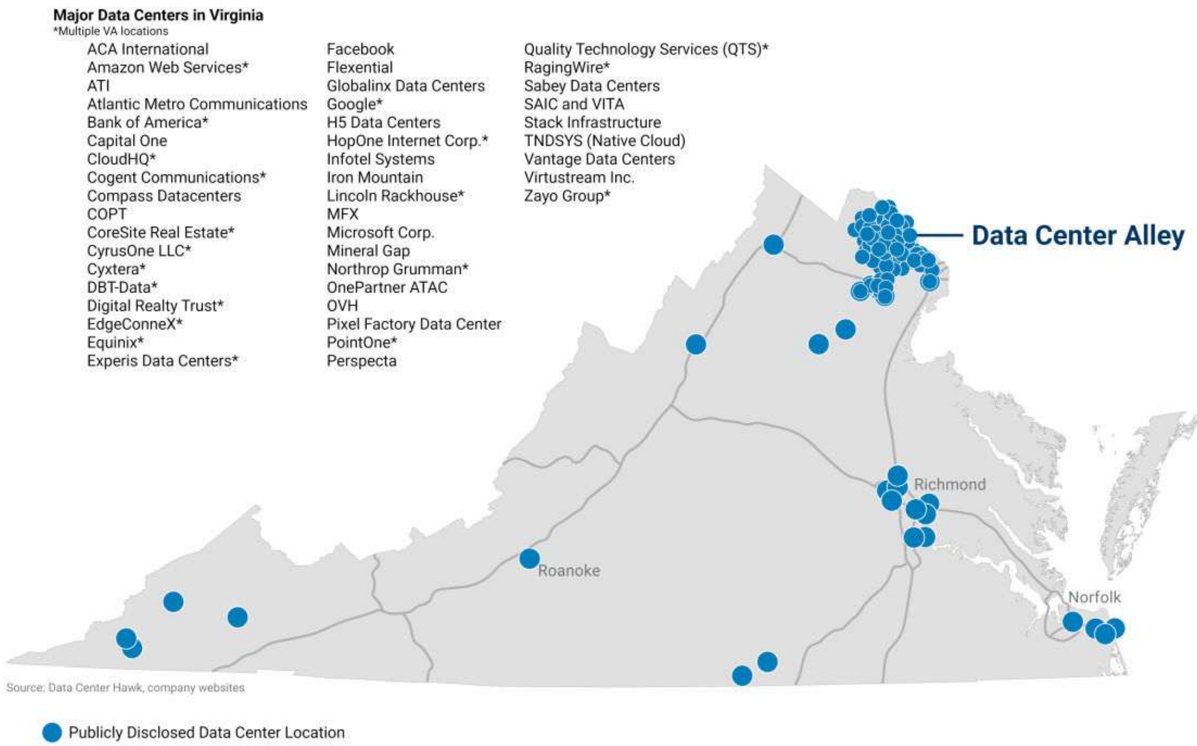
Power Usage Effectiveness (PUE)

PUE = Total Power / IT Power
Perfect = 1.0 (all to compute)
Typical: 1.5-2.0 | Hyperscalers: 1.1-1.2

Policy insight: Data center locations have environmental implication, power & water consumption, community externalities.

Exploring Cloud Infrastructure

[Cloud Infrastructure Map]



Data Center Alley

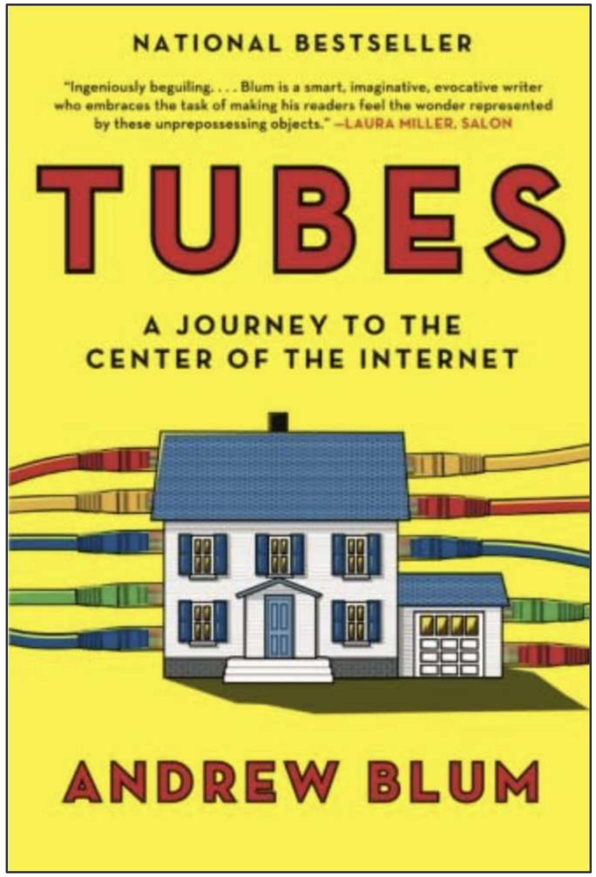
Northern Virginia hosts the world's densest concentration of data centers.

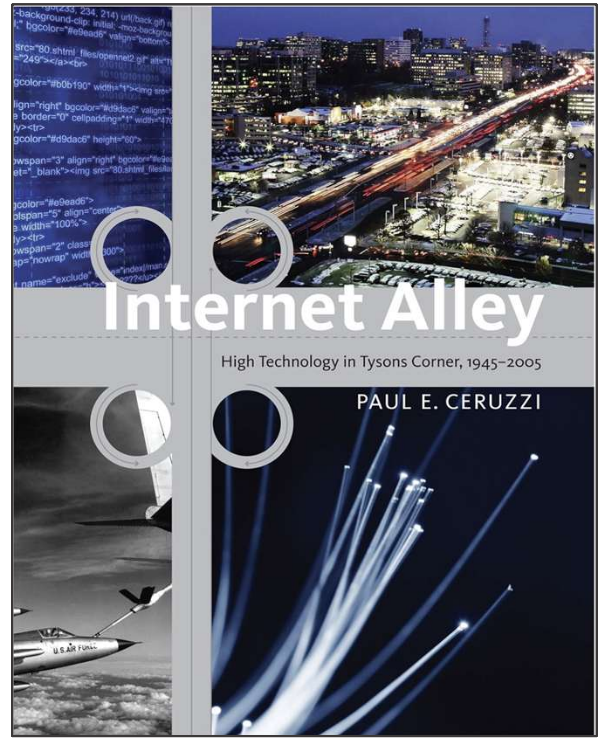
AWS us-east-1 is here.

Why here? MAE-East (early internet exchange), federal customers, established fiber

Operators: AWS, Microsoft, Google, Equinix, Digital Realty, QTS, dozens more

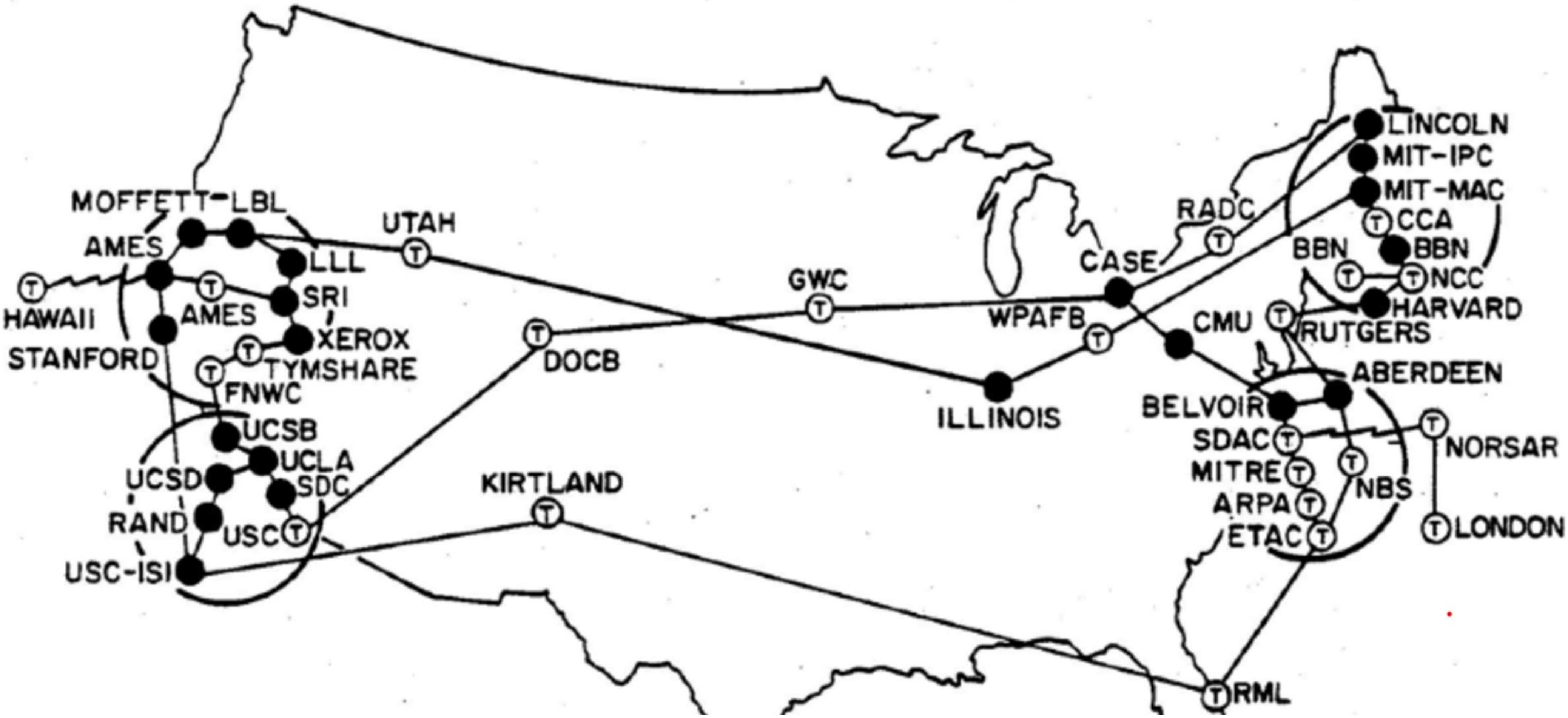
Two Recommended Sources





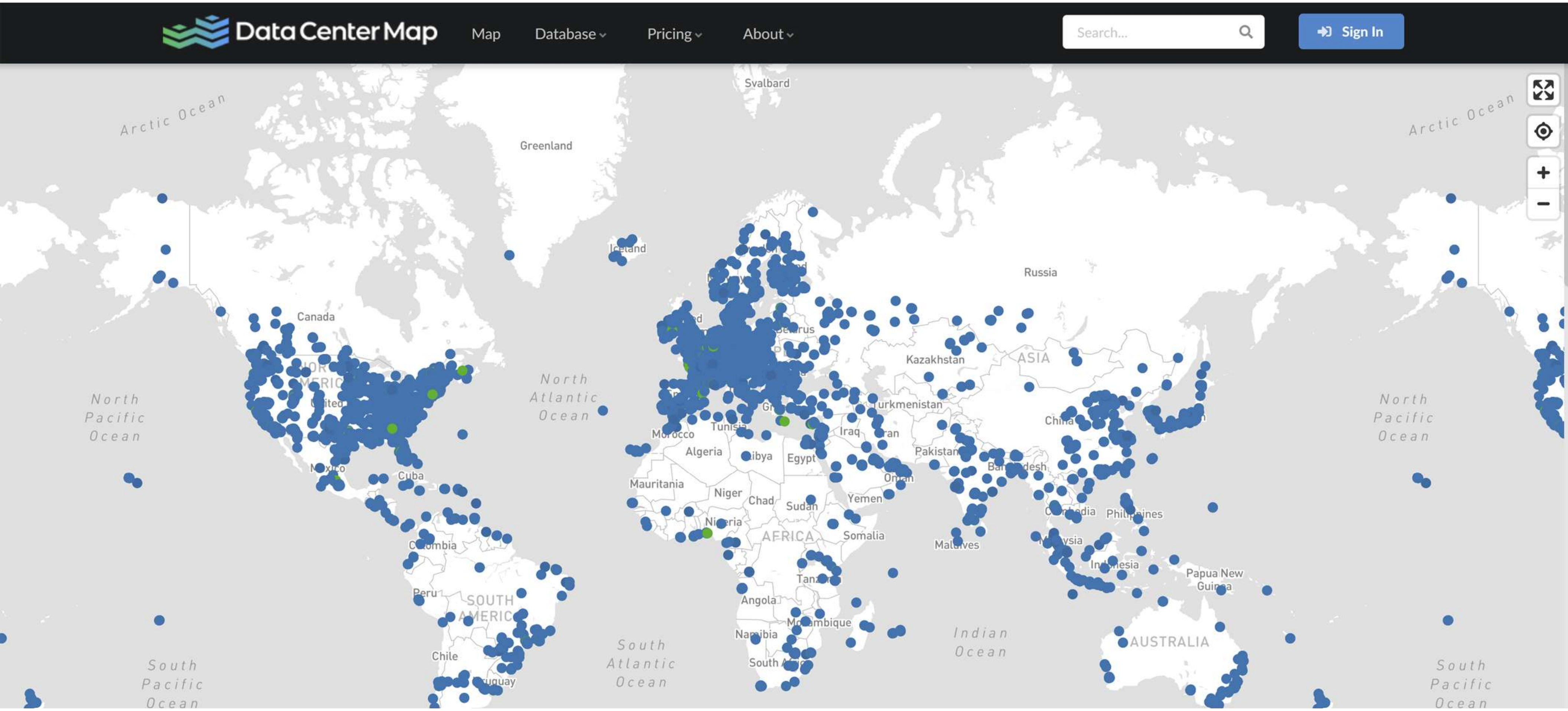
Across Data Centers

Regions, AZs, and the Physical Internet



Arpanet

[ARPANET Technical Information: Packet Formats, Maps, etc]



Data Center Map - 2025

[Data Center Map - Colocation, Cloud and Connectivity]

Availability Zones

What They Are

Physically separate facilities within a metro region
Independent power, cooling, network
Connected by high-bandwidth, low-latency links

Why They Exist

Protect against building-level failures
Designing for failure principle
Limit blast radius of any incident

Same principle as RAID at disk level: distribute across independent failure domains.

Above AZs: Regions

1. Latency to Users

<100ms: feels instant
100-300ms: noticeable
Tokyo→Virginia: ~150ms (physics)

2. Data Sovereignty

GDPR transfer restrictions
Data localization laws
HIPAA, FedRAMP, etc.

3. Disaster Recovery

Natural disasters affect regions
Regional outages happen
Multi-region = complex but critical

Factor	Questions to Ask
User location	Where are your users? What latency is acceptable?
Compliance	What regulations apply? Where must data reside?
Services	Are the services you need available in this region?
Pricing	Costs vary by region, have you compared?
Disaster recovery	What is your paired region strategy?

The Big Three Cloud Providers

Provider	Services	AZs	Regions	Share
AWS	>250	108	34	~31%
Microsoft Azure	>200	126	64	~25%
Google Cloud	>200	121	40	~11%

Also: Alibaba (China), Oracle, IBM | **Special:** GovCloud (AWS/Azure), DoD regions

The Physical Internet: Undersea Cables

95%+ of intercontinental data travels through undersea fiber

~400+ cables worldwide
Modern: 12-24 fiber pairs, 200+ Tbps

$200-500M per cable
~25 year lifespan

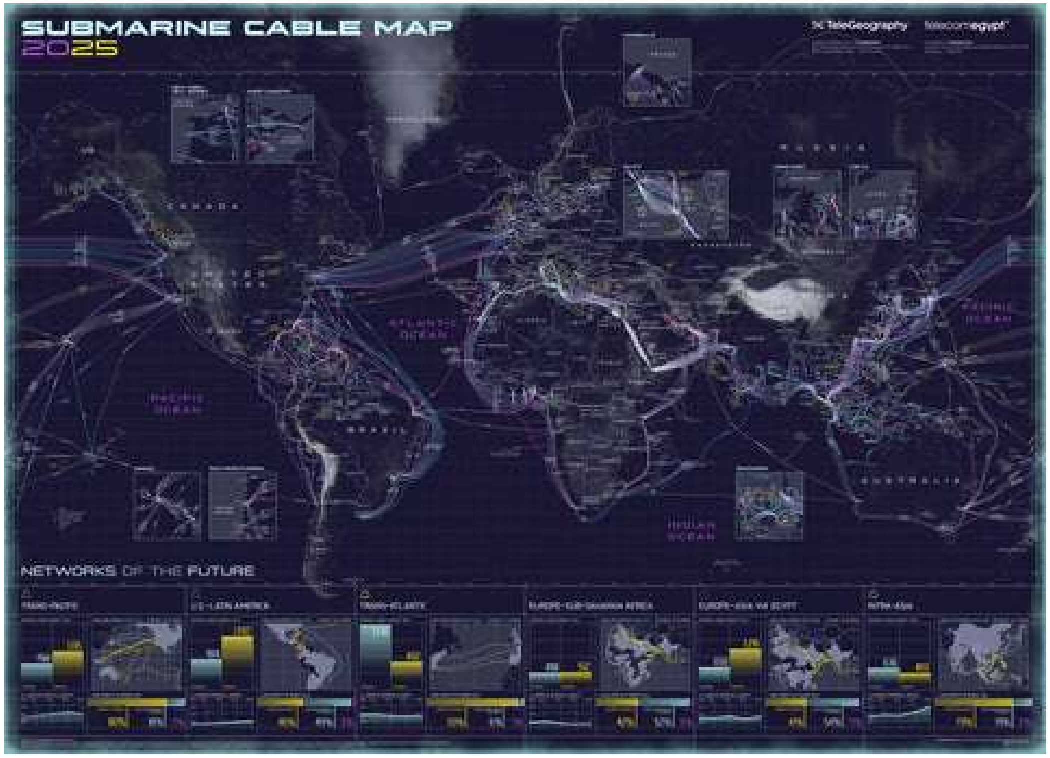
Owned by hyperscalers
Google, Meta, Microsoft, Amazon

Geopolitical chokepoints: Egypt/Red Sea, Singapore Strait, Cornwall UK, Miami | Critical infrastructure, few private owners

[Telegeography]

Undersea cables carry 95% of intercontinental data traffic

https://submarine-cable-map-2025.telegeography.com/



The Environmental Footprint

This is the standard content slide layout. Use it for:

· Explanatory content and concepts
· Key points and takeaways
· Supporting details and examples
· Lists of 3-5 items work best

Source: Citation goes here

Evaluating Sustainability Claims

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· Supporting details and examples
· Lists of 3-5 items work best

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Environmental Considerations

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· Supporting details and examples
· Lists of 3-5 items work best

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Country Briefing Applications

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· Supporting details and examples
· Lists of 3-5 items work best

Source: Citation goes here

Case Study: Region Selection Gone Wrong

SCENARIO

A U.S. agency consolidated all infrastructure in **us-east-1** (Virginia) for "simplicity and cost savings."

PROBLEMS THAT EMERGED

Alaska/Hawaii offices: 150ms+ latency—collaboration tools unusable

us-east-1 outage: entire agency lost access to all systems

Data sovereignty review: some partner data couldn't legally be stored in U.S.

The "simple" solution created operational, resilience, and compliance problems that cost more to fix than multi-region would have originally.

. Edge Computing & CDNs

Computer Level	Internet Equivalent	Function
L1 Cache	Edge Location	Tiny, ultra-fast cache closest to user (<50ms)
L3 Cache	Regional POP	Larger cache serving a metro area
RAM	Origin Shield	Intermediate cache protecting origin
Disk (HDD)	Origin Server	Authoritative source, highest latency

When to Use Edge Computing

- **Static content (images, video, CSS/JS):** Almost always cache at edge
- **Personalized content:** More complex—may require origin or edge compute
- **Real-time processing (IoT, video):** Often must process at edge—can't afford round-trip
- **Latency-critical (gaming, video calls):** Edge essential for acceptable experience

Source: Citation goes here

Judgment Question

DISCUSSION

Your agency serves users across the U.S. and **three European countries**. A contractor proposes consolidating all data in us-east-1 "for simplicity."

What concerns would you raise?

What additional information would you need?

Data Sovereignty Fundamentals

This is the standard content slide layout. Use it for:

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· Lists of 3-5 items work best

Source: Citation goes here

Sovereign Cloud Offerings

Offering	Key Features
AWS European Sovereign Cloud	Physically and logically separate from other AWS regions; located in Europe; operated by EU-resident AWS employees; designed for strictest data residency requirements
Microsoft EU Data Boundary	Commitment to store and process EU customer data within the EU; includes Azure, M365, Dynamics 365
Google Sovereign Controls	Customer-managed encryption keys; data residency controls; access transparency and approval
AWS GovCloud (U.S.)	Isolated regions for U.S. government workloads; FedRAMP High; operated by U.S. persons on U.S. soil

Source: Citation goes here

China Cloud

AWS China Gateway

Global-to-China: Extend your success with AWS in China

Contact Us For Support

Let AWS Help You Build a China-Ready Business

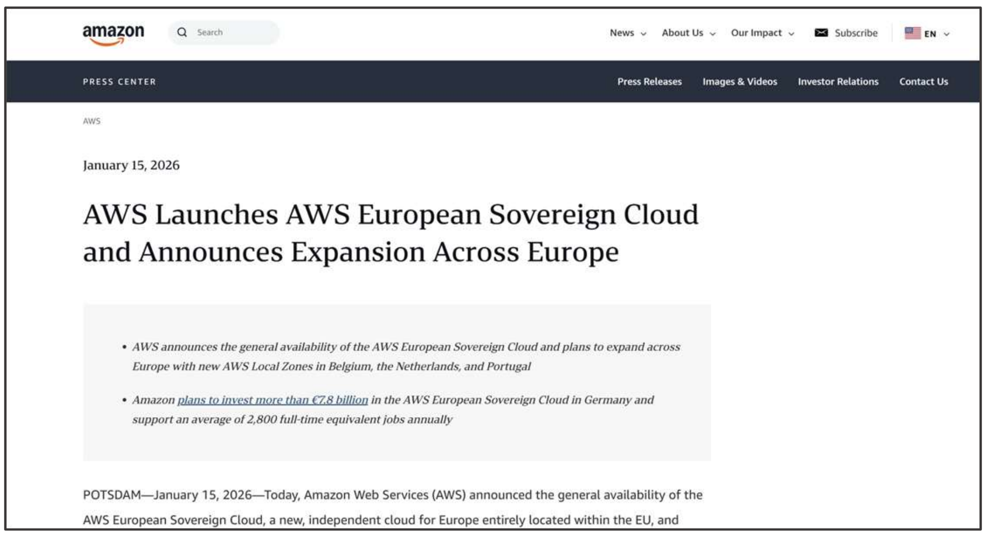
AWS helps millions of customers worldwide transform their business, enabling digital transformation that unlocks innovation and customer success. AWS China region started its

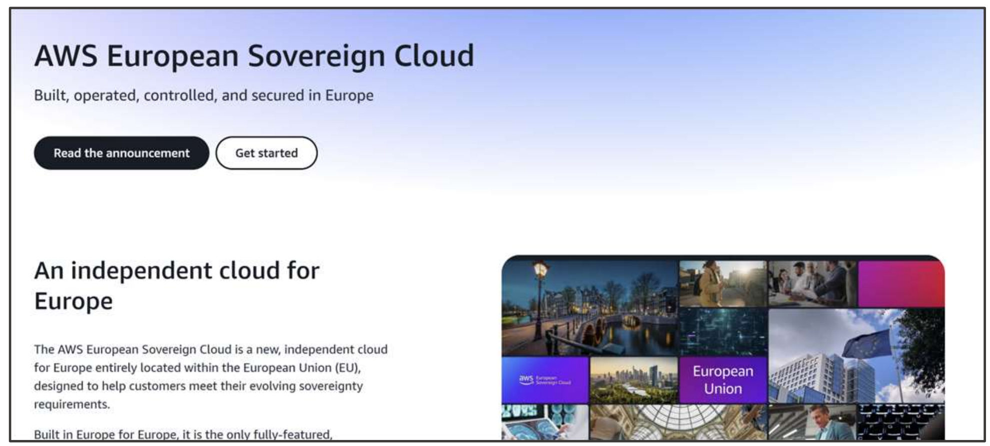
[AWS China Gateway - Power your business in China]()

AWS EU

"European Sovereign Cloud"

[AWS Launches AWS European Sovereign Cloud and Announces Expansion Across Europe - US Press Center]()





GovCloud & DoD Cloud

[Cloud for U.S. Defense | AWS]

CERN Case Study

How to Filter 50 TB/s to 10 GB/s

The Tiered Trigger System

Required reduction: **~5,000×**

Stage	Technology	Latency	In	Out	Reduction
Level-1 Trigger	Custom hardware, FPGAs	~4 μs	40 MHz	~100 kHz	400×
High-Level Trigger	Software (CPUs + GPUs)	~ms	100 kHz	~1-3 kHz	30-100×
Permanent Storage	Disk and tape	—	~1-3 kHz	—	—

Only ~0.003% of events are kept. These events led to discovering the Higgs boson.

The Design Philosophy

Latency vs. Sophistication

L1: fast but crude features

HLT: slow but full reconstruction

No Second Chances

If L1 rejects, it's gone forever

Edge decisions determine future questions

Specialized Hardware

When microseconds matter, use FPGAs

Same reason network switches use custom silicon

Domain Knowledge

Discovery came from 0.003%

Trigger was smart enough to keep the right events

Judgment Question

DISCUSSION

An agency wants to deploy **10,000 environmental sensors** generating readings every second. They ask whether to send all data to the cloud or process locally first.

What factors determine your recommendation?

What are they giving up with each choice?

AWS Services

What Cloud Providers Provide

Some of AWS Service Domains

Domain	Services	What They Do
Compute	EC2, Lambda, Fargate, EMR, Glue	Run code and process data
Storage	S3, Glacier, EBS, EFS	Store data at various tiers
Database	RDS, Aurora, DynamoDB, Redshift	Managed database services
Security	IAM, VPC, KMS	Access control, networking, encryption
Migration	DMS	Move databases to AWS

Free Tier Warning

AWS IS NOT FREE!

Most services have free tier (1 year limited usage)

Monitor usage carefully

Set up billing alerts

If you explore on your own: BE CAREFUL

Documentation: docs.aws.amazon.com

Key Takeaways

The Recurring Mental Models

1. Faster/Closer Is Always More Expensive — L1 cache costs more than RAM, hot storage more than cold, edge more than origin

2. Redundancy at Every Level — RAID→servers→AZs→regions; each adds cost and complexity

3. Abstraction Trades Control for Convenience — VMs simpler than physical, serverless simpler than VMs

4. Physical Constraints Govern Everything — Speed of light, heat, power costs—engineering manages, not eliminates

5. Edge Decisions Determine Future Questions — CERN's trigger, sensor networks—filtering constrains downstream

Mapping Concepts Across Scales

Concept	Computer	Data Center	Global
Fast/small cache	L1 cache	Local NVMe	CDN edge
Slow/large storage	HDD	Tape archive	Origin server
Bus/network	PCIe bus	Spine-leaf fabric	Undersea cables
Failure domain	Single disk	Single rack/AZ	Single region
Specialization	GPU	ML accelerator cluster	Specialized region

Your Role as Policy Professionals

You are not building these systems. You are:

Procuring	Overseeing	Governing	Deciding
Understanding what you're buying	Asking the right questions	Ensuring compliance	Recognizing trade-offs

The goal: Mental models to reason about infrastructure, evaluate vendor claims, and ask intelligent questions when told something is "impossible" or "easy."

Key Takeaways

1. Cloud technologies aren't just "someone else's computer"—they're carefully designed systems emerging from physics and economics

2. The same patterns repeat at every scale (the fractal)

3. Understanding why systems are organized this way lets you reason about systems you haven't seen

4. Your role as policy professionals is informed judgment, not technical execution

Resources & Further Reading

Interactive Tools

- submarinecablemap.com
- cloudping.info
- AWS/Azure/GCP Infrastructure Maps

Documentation

- AWS Well-Architected Framework
- NIST Cloud Definition
- CERN Computing docs

Books

- "Internet Alley" - Ceruzzi
- "Tubes" - Andrew Blum
- "The Datacenter as a Computer" - Google

Policy-Relevant

- datacenterdynamics.com
- CFR: Submarine Cables reports
- EU Digital Sovereignty initiatives

Questions?

Computer to Cloud Computing

PPOL 5206: Massive Data Fundamentals